

NDN in IIoT

Data-driven communications in Industrial Environments

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NDN Community Event 2021 (remote talk)
Panel: NDN beyond NDN: Data-driven applications and services

About fortiss

Research Institute of the Free-state of Bavaria for Software
Intensive Systems – Associated to the Technical University of
Munich

<http://www.fortiss.org>

- **Legal form:** non-commercial limited company
- **Foundation:** 2009
- Shareholders: Free State of Bavaria (2/3) and Fraunhofer-Gesellschaft (1/3)
- 2 locations in Munich, 173 researchers

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Industrial IoT @fortiss

Current Research Lines

Decentralised Edge Computing

- Context-aware Cloud to Edge continuum - Smart, Context-aware Edge selection
- Automated Service-to-Thing matching
- Brownfield devices, semantic integration into IoT systems

Real-time IoT System Adaptation

- Dynamic, context-aware micro-service management
- Deterministic guarantees for time-sensitive traffic across wired/wireless networks

In-network computing

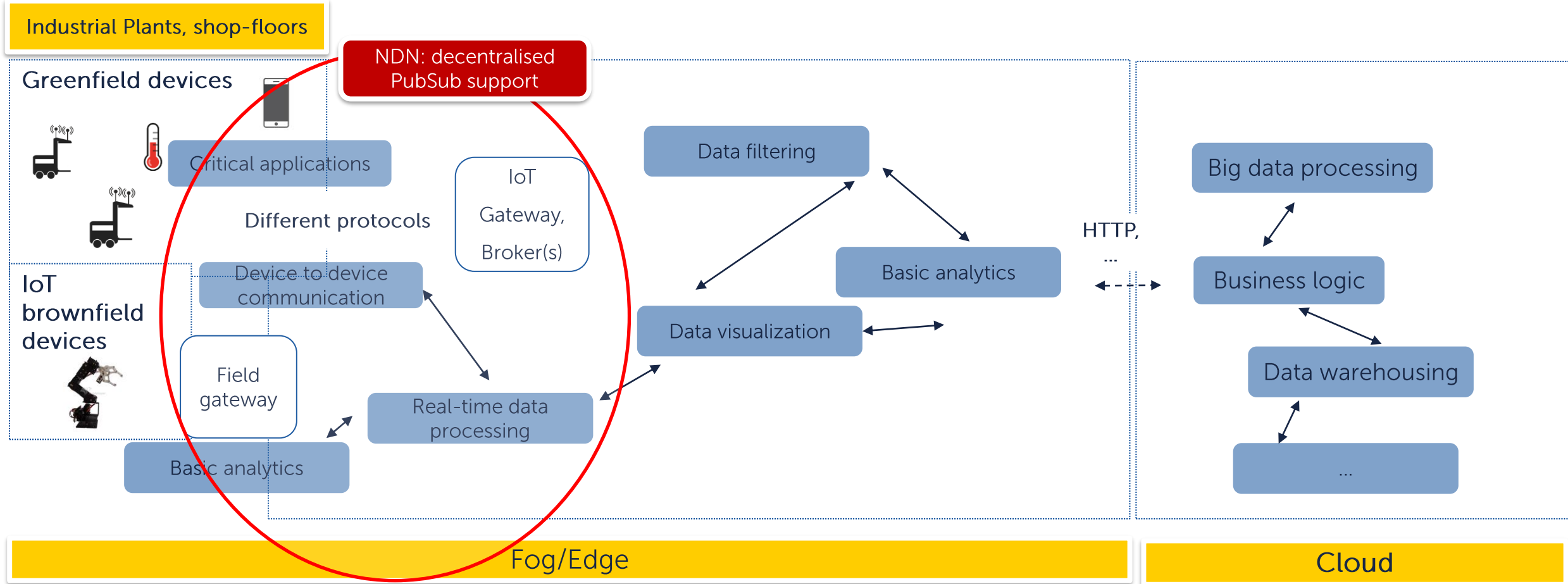
- Addressing the convergence of network, compute, data
- ICN based PubSub Data exchange (device to Edge, Edge to Edge)
- SDN in IIoT



Bringing Computation, Network and Data Together

Industrial IoT Example

Containerized micro-service



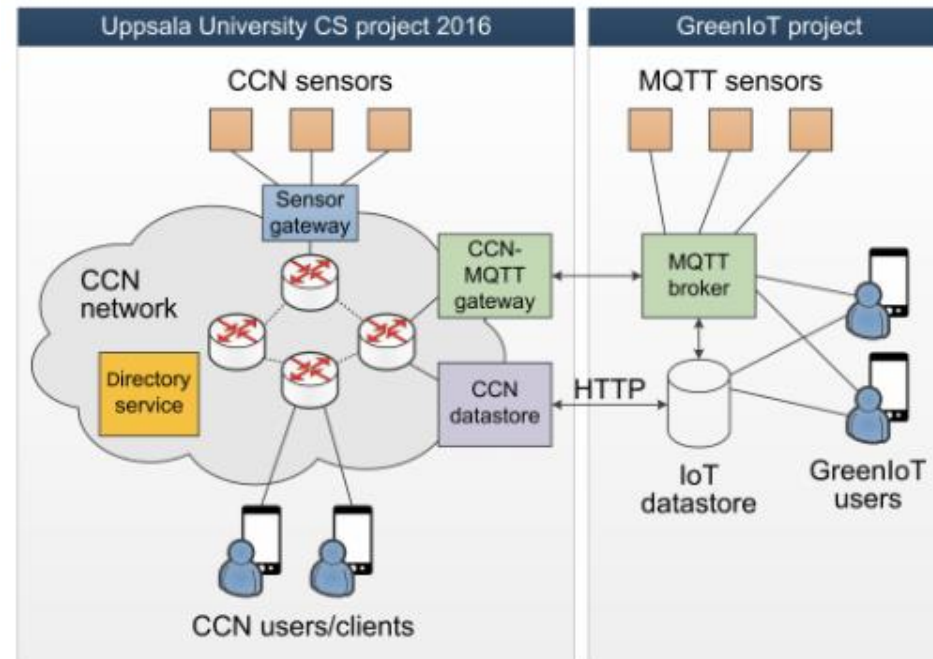
Revising NDN for IIoT Challenges

NDN has, by design, interesting features for IoT

- Integrated security
- In-network caching
- Decentralisation
- Flexible forwarding strategies
- Interface abstraction (interfaces, services, applications)

For IIoT, several challenges:

- in-plant communication requires tight control loops
- Static environments
- 1-hop usually
- **Interoperability is required**



*Bengt Ahlgren (SICS), "ProjectGreenIoT: An Energy-Efficient IoT Platform for Open Data and Sustainable Development" (2018).

Revising NDN for IIoT

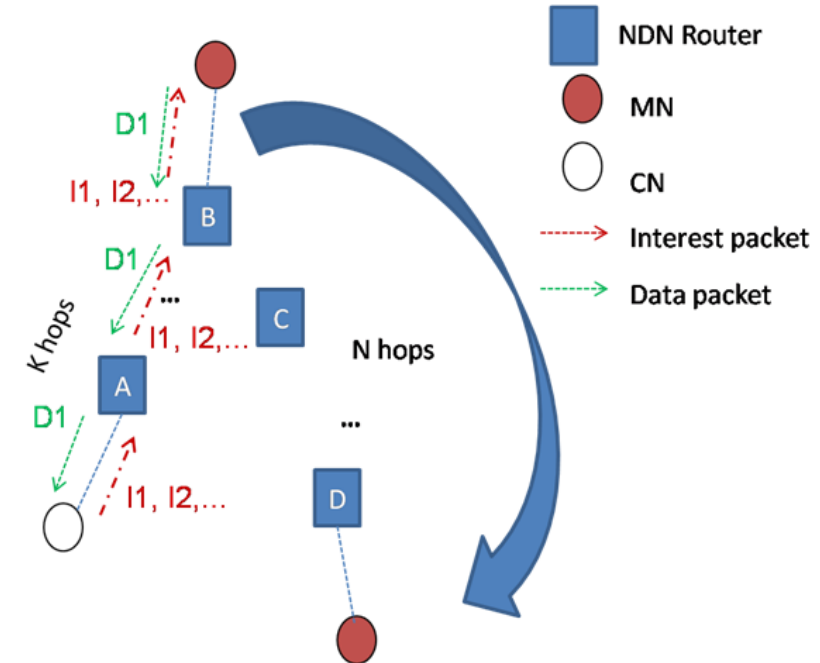
Challenges I

MOBILITY (not critical yet but: needs to be addressed)

- Producer and consumer should be handled equally from a mobility management perspective
- Mobility anticipation mechanisms can reduce signaling and improve producer mobility support
- Naming in ICN is hierarchical and independent of location.
 - Still, the common way to handle naming is to associate with routing domains, e.g. “/bla.de/videos/”. This may affect routing
 - Middleware would benefit from naming guidelines

IN NETWORK CACHING

- Currently, strategies are based on fixed networks
- Need to take into consideration the temporary nature of IoT data (data transiency)



Revising NDN for IIoT

Challenges II

NAMING

- Applications select naming schemes – naming is opaque to routing
- Hierarchical and flat schemes are possible
- NDN adopts 1-dimensional hierarchical naming and lookup operations based on longest-prefix matching
 - Multi-dimensional naming is relevant (location, period scopes)
 - Multi-dimensional naming translation into longest-prefix matching (for routing) is under development
- No consensus on semantics adoption
 - Richer semantics may bring in security issues (e.g., allow leakages of personal information)
 - Obfuscation must not endanger other NDN functionality, e.g., routing

PULL AND PUSH BASED COMMUNICATIONS*

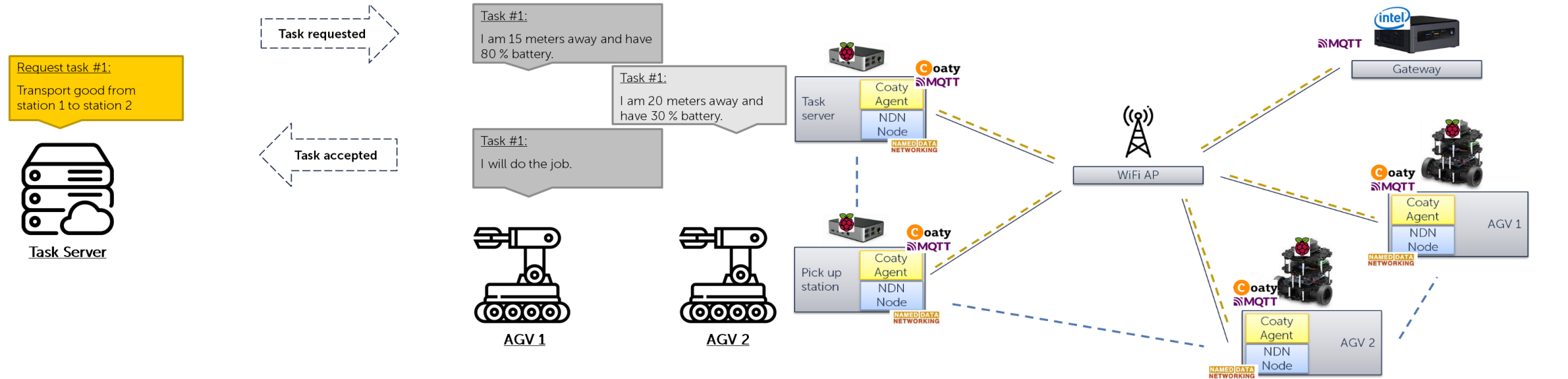
- Pull communication is not enough to support even-triggered or periodic communications
- Push based models need to be adjusted to serve IoT requirements

MULTI-PARTY DATA SYNCHRONIZATION

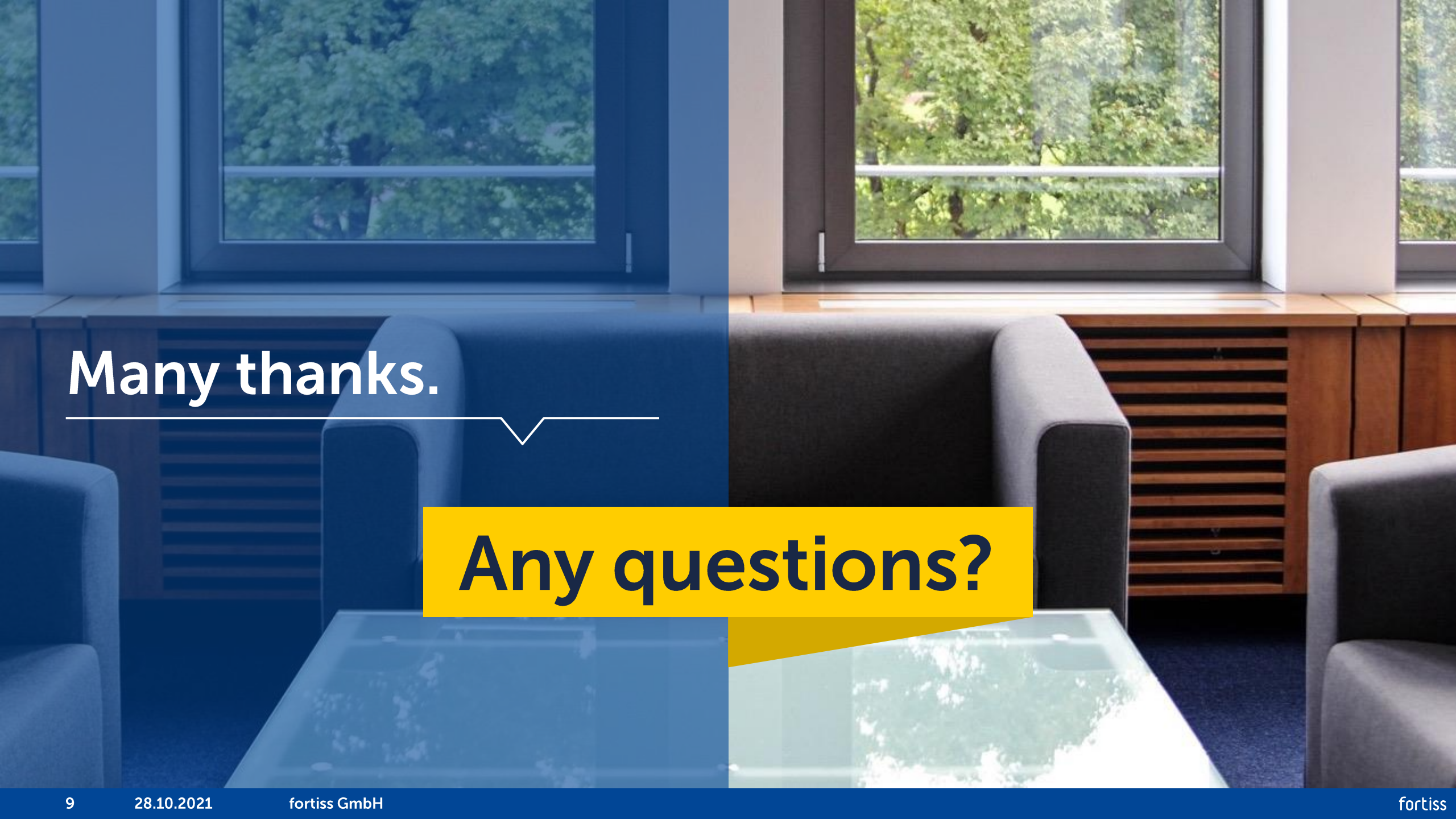
- NDN supports N to M communication
 - Guarantees delivery even if connectivity is intermittent
 - Does not guarantee synchronization among consumers
- Data set synchronization is required, for instance, in Smart Health or Connected Vehicles scenarios
- A few options available: Chronosync, VectorSync (abstractions between middleware and network primitives)

NDN in IIoT

Demonstrator, decentralised data exchange in IIoT scenarios



- Comparison to MQTT (broker approach) and to OPC UA (PubSub, multicast)
- Comparison to the coaty.io framework (discovery, multi-party data synchronisation, "decentralised" data exchange)
- Energy consumption: network and node
- Latency: time to completion (request from pick up station to AGV)
- Messaging overhead: messages during registration and runtime
- Node usage: memory, CPU, etc.



Many thanks.

Any questions?

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